

HERA-PdM: Lessons from Deploying Uncertainty-Gated Edge AI for Predictive Maintenance, from Bearings to a Truck Fleet

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Abstract: Putting predictive-maintenance (PdM) models on a factory floor raises questions that accuracy scores leave open: where inference runs, how far a prediction can be trusted, and when a person decides. We study them with HERA-PdM. Cross-validated conformalized quantile regression (CQR-CV+) turns remaining-useful-life (RUL) predictions into intervals, the edge escalates only when its interval does not settle the maintenance action, and costly actions go to a human as SHAP-backed recommendations. On 17 FEMTO and 15 XJTU-SY run-to-failure bearings, CQR-CV+ raised coverage to 0.94–0.96 and 0.93–0.99 (nominal 0.90), and calibrated decisions cut critical snapshots left without action from 49.5% to 0.6% (FEMTO) and 34.2% to 9.2% (XJTU-SY), at the cost of more false maintenance. Escalation helped only under gradual degradation: on XJTU-SY, escalating 79.1% of snapshots matched always-cloud safety with less false maintenance (53.2% vs. 69.1%), though not for every bearing. Our gate did no better than a plain width gate, and per-bearing alarms came too early because a dozen calibration units are too few for risk control. On 23,550 Scania trucks, the same risk control met its lateness target (9.9% vs. 40.5% for a fixed threshold) and HERA escalated 73% of readouts at always-cloud safety, but false alarms stayed high.

Index Terms: AI deployment, predictive maintenance, edge AI, conformal prediction, explainable AI, human-in-the-loop, Digital Twin

Key findings. Across two bearing datasets with very different degradation and a truck fleet, (i) calibration removes most unsafe point-threshold decisions; (ii) escalation pays off only when degradation shows early; (iii) interval width matters more than the gating rule; (iv) per-snapshot guarantees do not give timely per-bearing alarms; and (v) unit-level lateness guarantees need fleets, not a dozen bearings.

Our contributions are:

- 1) HERA-PdM, a reproducible four-layer framework whose *decision-sufficiency* gate escalates only when the edge RUL interval straddles a maintenance-decision boundary, which bounds locally resolved unsafe decisions by the edge miscoverage level.
- 2) A bearing-grouped, two-dataset benchmark of ten tuned quantile models with and without DT context, showing that CQR-CV+ restores nominal coverage and that larger fog/cloud models help only under gradual degradation.
- 3) Snapshot-, unit- and fleet-level evidence with bootstrap intervals, and a feasibility bound for unit-level risk control (Proposition 1) checked on 23,550 trucks, distilled into deployment lessons.

I. INTRODUCTION

Predictive maintenance (PdM) estimates the remaining useful life (RUL) of a machine from condition data so that repairs can be planned before failure [1]. Running it on a real factory floor raises three questions that accuracy alone does not answer. *Where* should inference run? Vibration is sensed at the machine, and edge–cloud architectures keep routine inference local [2]. *How far* can a prediction be trusted? That calls for calibrated uncertainty, not point estimates [3]. And *who* acts on it? Industry 5.0 keeps people in charge of consequential decisions [4].

Hierarchical PdM systems combining edge devices, Digital Twins (DTs) and agents already exist [5], [6]. They escalate from edge to cloud on anomaly thresholds [5] or uncalibrated confidence [6]. Calibrated deferral with guarantees exists for classification cascades [7], but not for RUL-based maintenance. We therefore ask *when* conformally calibrated RUL intervals computed on the edge are a useful escalation signal, and what they cost in safety, false maintenance, alarm timing, communication and latency.

II. RELATED WORK AND RESEARCH GAP

RUL prediction. Deep sequence models such as temporal convolutional networks [8] are now standard RUL estimators. Reviews [1] treat health-stage division, that is, detecting when degradation first becomes observable, as a prerequisite for prognosis. Our datasets [9], [10] differ sharply here. *Hierarchical PdM.* Ren et al. [2] run a lightweight TCN on the edge and refine its estimate in the cloud. HAMA [5] escalates from an edge z-score pre-filter to fog ensembles with SHAP and generated operator text. EC-Distill-ZeroDiag [6] hands uncertain fault-diagnosis cases from an edge agent to a cloud model. *Calibrated uncertainty.* CQR [3] and CV+ [11] give coverage without distributional assumptions. Conformal RUL intervals have been built for bearings [12], but not tied to a decision or routing layer. Conformal risk control [13] bounds expected monotone losses, and conformal alignment certifies edge–cloud cascades for classification [7].

Gap. We coded 42 papers (Table I). None uses conformal RUL intervals computed on the edge as the escalation rule of

TABLE I
CLOSEST RELATED WORK (CODED FROM FULL TEXT, ABSTRACTS OR PUBLIC CODE). ✓: PRESENT, ○: PARTIAL, -: ABSENT.

Work	DT	Edge	Agents	RUL	Calib.	UQ	Routing	XAI	HIL
Ren et al. [2]	-	✓	-	✓	-	-	-	-	-
HAMA [5]	-	✓	✓	○	-	-	○ anomaly	✓	○
EC-Distill [6]	-	✓	✓	-	-	-	✓ conf.	○	-
CAB cascade [7]	-	✓	-	-	✓	-	✓	-	-
Wang et al. [12]	-	-	-	✓	✓	-	-	-	-
HERA-PdM	○	✓	✓	✓	✓	✓	✓	✓	✓

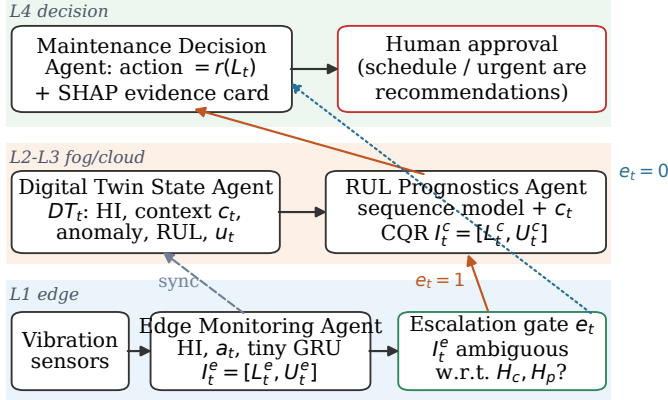


Fig. 1. HERA-PdM. The edge agent computes a calibrated RUL interval I_t^e ; the gate escalates only if I_t^e is not decision-sufficient; the fog/cloud agent uses the DT context c_t ; schedule/urgent actions need human approval.

a hierarchical PdM system, links escalation to maintenance-decision boundaries, and compares it with the gates of existing systems on public run-to-failure data from different degradation regimes. The architecture itself is not new; our contribution lies in the method and its evaluation.

III. HERA-PdM METHODOLOGY

Fig. 1 shows the four layers. The “agents” are software components with fixed responsibilities and messages; no language model is involved.

Edge Monitoring Agent. Each vibration snapshot is reduced to 28 features $x_t \in \mathbb{R}^{28}$ (per axis: RMS, kurtosis, skewness, peak-to-peak, crest, impulse and shape factors, spectral centroid and six log band energies). They are z-scored against the asset’s first snapshots and compressed by $\text{sign}(z) \log(1+|z|)$, giving the health index $h_t = \log(1 + \frac{1}{d} \sum_j |z_{t,j}|)$. The anomaly score a_t , the smoothed h_t standardized by the asset’s raw baseline, alarms above τ_a , the 99th percentile of healthy training snapshots. A tiny model maps a short window to three RUL quantiles.

Digital Twin State Agent. The twin is a synchronized state record $DT_t = \{\text{condition, sensor state, } h_t, \text{ context } c_t, \hat{R}_t, I_t, u_t, \text{ anomaly state}\}$, updated by a 12-byte sync (t, h_t, a_t) after each snapshot. The context $c_t \in \mathbb{R}^6$ holds fast and slow averages of h_t , their difference, its running maximum, its slope and the cumulative excess over baseline. Absolute operating time is left out because lifetimes differ by an order of magnitude.

RUL Prognostics Agent. A sequence model g_ϕ takes a longer window X_t together with c_t and outputs non-crossing quantiles $(\hat{q}_t^{\alpha/2}, \hat{q}_t^{0.5}, \hat{q}_t^{1-\alpha/2})$, trained with the pinball loss on the capped target $y_t = \min(\text{RUL}_t, R_{\max})$.

Calibration (CQR-CV+). We split the non-test assets into $K=4$ groups balanced by condition and train an inner model $g^{(-k)}$ without group k . This yields an out-of-fold score $s_i = \max(\hat{q}_i^{\alpha/2} - y_i, y_i - \hat{q}_i^{1-\alpha/2})$ for every non-test snapshot i in group $k(i)$. Following CV+ [11] with CQR scores [3],

$$L_t = Q_\alpha^- \{\hat{q}_t^{\alpha/2, (-k(i))} - s_i\}, \quad (1)$$

$$U_t = Q_{1-\alpha}^+ \{\hat{q}_t^{1-\alpha/2, (-k(i))} + s_i\}, \quad (2)$$

where Q_α^- and $Q_{1-\alpha}^+$ are the $\lfloor \alpha(n+1) \rfloor$ -th and $\lceil (1-\alpha)(n+1) \rceil$ -th smallest values over the n calibration snapshots, and $\hat{R}_t$ is the mean inner median. For exchangeable assets CV+ guarantees coverage $\geq 1 - 2\alpha$ (typically $\approx 1 - \alpha$); snapshots of one asset are dependent, so the guarantee is approximate and we test it. The baseline calibrator is split CQR on three held-out bearings.

Decision and escalation. Critical and planning horizons H_c and H_p define the actions: $r(v) = \text{URGENT}$ if $v \leq H_c$, SCHEDULE if $H_c < v \leq H_p$, and CONTINUE otherwise. The Maintenance Decision Agent acts on the lower bound, $\pi_t = r(L_t)$, and turns CONTINUE into INSPECT when $a_t > \tau_a$. SCHEDULE and URGENT are recommendations that need a person’s approval and come with an explanation card (RUL, interval, anomaly evidence, top-3 SHAP [14] contributors, rationale). With $B = \{H_c, H_p\}$, the edge escalates when its interval is ambiguous or disagrees with the anomaly evidence:

$$e_t = \mathbb{I}[\exists h \in B: L_t^e \leq h < U_t^e] \vee \mathbb{I}[a_t > \tau_a \wedge r(L_t^e) = \text{C}], \quad (3)$$

where C stands for CONTINUE . A local CONTINUE requires $L_t^e > H_p \geq H_c$, so a critical asset ($y_t \leq H_c$) keeps running without escalation only if $y_t \notin I_t^e$:

$$\Pr(e_t=0, \pi_t=\text{CONTINUE}, y_t \leq H_c) \leq \Pr(y_t \notin I_t^e). \quad (4)$$

False local URGENT actions are bounded in the same way. Unlike a width threshold, the gate ignores wide intervals that sit inside one action region and escalates narrow ones that cross a boundary.

Cost model. We count payloads in float32. Always-cloud streams x_t (112 B per snapshot); HERA sends the 12 B sync, plus the untransmitted part of the window when $e_t=1$. Latency is the edge time plus, on escalation, the RTT and the cloud time, using host timings (Sec. V) and an assumed RTT of 50 ms.

IV. EXPERIMENTAL SETUP

Data. FEMTO-ST/PRONOSTIA [9]: 17 bearings under three conditions, 0.1 s snapshots every 10 s (24,889 snapshots), full test trajectories; several fail abruptly. XJTU-SY [10]: 15 bearings at 35 Hz/12 kN, 37.5 Hz/11 kN and 40 Hz/10 kN, 1.28 s snapshots every minute (9,216 snapshots); most degrade gradually. RUL is capped at $R_{\max}=3000$ s (FEMTO) and 6000 s (XJTU-SY), with $H_c=0.2R_{\max}$ and $H_p=0.6R_{\max}$;

TABLE II

RUL ACCURACY AND CALIBRATION ON HELD-OUT BEARINGS (MEAN $\pm$ STD OVER 3 SEEDS; NOMINAL COVERAGE 0.90). RAW: UNCALIBRATED QUANTILES; SPLIT: CQR CALIBRATED ON 3 HELD-OUT BEARINGS PER FOLD; CV+: CQR-CV+ CALIBRATED ON ALL NON-TEST BEARINGS. MARKS: ARCHITECTURE SELECTED ON INNER-VALIDATION PINBALL LOSS FOR FEMTO ($\dagger$) AND XJTU-SY (*).

Model	FEMTO/PRONOSTIA (17 bearings, $R_{\max}=3000$ s)					XJTU-SY (15 bearings, $R_{\max}=6000$ s)				
	RMSE (s)	MAE (s)	Raw cov.	Cov.	MPIW (s)	RMSE (s)	MAE (s)	Raw cov.	Cov.	MPIW (s)
Edge-CNN* (split)	1114 $\pm$ 40	851 $\pm$ 19	0.37	0.88 $\pm$ 0.02	2080	2278 $\pm$ 389	1859 $\pm$ 330	0.65	0.97 $\pm$ 0.00	3716
Edge-GRU $\dagger$ (split)	820 $\pm$ 34	518 $\pm$ 62	0.71	0.84 $\pm$ 0.04	1877	2531 $\pm$ 642	2004 $\pm$ 601	0.85	0.87 $\pm$ 0.08	4184
LSTM (split)	1208 $\pm$ 118	772 $\pm$ 102	0.50	0.85 $\pm$ 0.04	2263	1516 $\pm$ 127	886 $\pm$ 104	0.90	0.91 $\pm$ 0.01	2436
LSTM+DT $\dagger$ (split)	1124 $\pm$ 48	751 $\pm$ 54	0.37	0.84 $\pm$ 0.04	2154	1698 $\pm$ 58	1130 $\pm$ 68	0.89	0.88 $\pm$ 0.03	3107
GRU (split)	1171 $\pm$ 114	751 $\pm$ 84	0.40	0.86 $\pm$ 0.03	2151	1516 $\pm$ 39	1027 $\pm$ 73	0.89	0.89 $\pm$ 0.02	2404
GRU+DT* (split)	1140 $\pm$ 89	771 $\pm$ 66	0.32	0.89 $\pm$ 0.03	2286	1647 $\pm$ 94	1226 $\pm$ 153	0.86	0.83 $\pm$ 0.06	3042
TCN (split)	1159 $\pm$ 70	827 $\pm$ 72	0.16	0.85 $\pm$ 0.02	2241	1666 $\pm$ 119	1303 $\pm$ 145	0.52	0.95 $\pm$ 0.01	3691
TCN+DT (split)	1153 $\pm$ 136	855 $\pm$ 115	0.16	0.89 $\pm$ 0.01	2456	1496 $\pm$ 118	1100 $\pm$ 138	0.62	0.81 $\pm$ 0.11	3477
CNN-LSTM (split)	1106 $\pm$ 74	674 $\pm$ 34	0.40	0.85 $\pm$ 0.02	2119	1335 $\pm$ 3	772 $\pm$ 21	0.85	0.90 $\pm$ 0.01	2416
CNN-LSTM+DT (split)	1108 $\pm$ 39	747 $\pm$ 54	0.28	0.89 $\pm$ 0.01	2355	1446 $\pm$ 30	895 $\pm$ 33	0.84	0.91 $\pm$ 0.00	2686
Selected edge (CV+)	753 $\pm$ 5	501 $\pm$ 22	0.75	0.94 $\pm$ 0.01	2193	1938 $\pm$ 251	1677 $\pm$ 232	0.69	0.99 $\pm$ 0.01	4134
Selected cloud+DT (CV+)	892 $\pm$ 27	665 $\pm$ 14	0.28	0.96 $\pm$ 0.02	2506	1422 $\pm$ 38	1077 $\pm$ 59	0.91	0.93 $\pm$ 0.01	3030
Selected cloud, no DT (CV+)	954 $\pm$ 16	653 $\pm$ 13	0.35	0.95 $\pm$ 0.02	2551	1305 $\pm$ 54	757 $\pm$ 66	0.90	0.94 $\pm$ 0.01	2337

TABLE III

HIERARCHICAL POLICIES, ABLATIONS AND SYSTEM COST (MEAN $\pm$ STD OVER 3 SEEDS, ALL HELD-OUT BEARINGS). ESC.: ESCALATED SNAPSHOTS; UNSAFE: CRITICAL SNAPSHOTS ($y \leq H_c$) LEFT WITHOUT A MAINTENANCE ACTION; FM: FALSE MAINTENANCE ON HEALTHY SNAPSHOTS ($y > H_p$); B/S: PAYLOAD BYTES PER SNAPSHOT (ANALYTIC). $\dagger$ ESCALATION BUDGET MATCHED TO HERA ON CALIBRATION DATA.

Policy	FEMTO/PRONOSTIA					XJTU-SY				
	Esc. (%)	Unsafe (%)	FM (%)	RMSE (s)	B/s	Esc. (%)	Unsafe (%)	FM (%)	RMSE (s)	B/s
Always-edge	0.0	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	752	0	0.0	0.0 $\pm$ 0.0	77.1 $\pm$ 7.1	1485	0
Always-cloud	100.0	0.0 $\pm$ 0.0	96.9 $\pm$ 2.7	891	112	100.0	9.2 $\pm$ 1.7	69.1 $\pm$ 6.1	1421	112
Cloud point + fixed thr.	100.0	49.5 $\pm$ 2.7	21.7 $\pm$ 6.8	1090	112	100.0	34.2 $\pm$ 9.2	8.3 $\pm$ 2.1	1689	112
HERA-full	97.1	0.6 $\pm$ 1.1	95.0 $\pm$ 0.5	892	123	79.1	9.2 $\pm$ 1.7	53.2 $\pm$ 13.3	1381	104
HERA w/o anomaly term	97.1	0.6 $\pm$ 1.1	95.0 $\pm$ 0.5	892	123	79.1	9.2 $\pm$ 1.7	53.2 $\pm$ 13.3	1381	104
HERA-no-DT	97.1	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	953	123	79.1	12.1 $\pm$ 4.3	25.5 $\pm$ 7.7	1303	104
HERA-no-UQ (point margin)	48.1	56.2 $\pm$ 5.3	15.1 $\pm$ 2.8	989	89	67.8	34.2 $\pm$ 9.2	8.0 $\pm$ 2.2	2062	91
HERA-fixed (anomaly trig.)	3.9	0.6 $\pm$ 1.1	96.9 $\pm$ 1.4	760	27	5.4	0.0 $\pm$ 0.0	77.1 $\pm$ 7.1	1541	20
Width gate $\dagger$	97.2	0.6 $\pm$ 1.1	95.1 $\pm$ 0.5	892	123	79.2	9.2 $\pm$ 1.7	53.3 $\pm$ 13.3	1381	104
Linear score gate $\dagger$	94.3	0.6 $\pm$ 1.1	95.7 $\pm$ 0.9	892	121	77.9	9.2 $\pm$ 1.7	53.8 $\pm$ 13.3	1378	103
Random gate $\dagger$	96.4	0.0 $\pm$ 0.0	96.8 $\pm$ 2.6	887	124	71.6	6.5 $\pm$ 1.4	71.4 $\pm$ 6.3	1441	124

windows are 16/64 and 8/32 snapshots (edge/cloud). SCANIA Component X [15] (DOI 10.5878/bnh5-ka77, CC BY 4.0) has 23,550 trucks with irregular operational readouts (counters and histograms); 2,272 had a repair during the study and the rest are right-censored. We use it only for unit-level alarm timing (Sec. V).

Protocol. Bearing-grouped folds balanced by condition (4 for FEMTO, 5 for XJTU-SY) test every bearing once. Hyperparameters (width 32–128, 8 or 25 epochs, input noise) and the edge/cloud model choice were selected once, with seed 0, on two inner validation bearings held out from the training bearings of two folds (FEMTO: Edge-GRU/LSTM+DT; XJTU-SY: Edge-CNN/GRU+DT). Fold, calibration and CV+ group assignments are fixed lists shared by all three seeds, which change only the training randomness. Metrics are pooled over test bearings (mean $\pm$ std). Two-sided Wilcoxon signed-rank tests pair HERA and a comparator per bearing (seeds averaged, zero differences dropped).

Baselines and metrics. Models are LSTM, GRU, TCN [8]

and CNN-LSTM with and without DT context, plus MSE point models. Policies are always-edge, always-cloud and point-with-fixed-threshold; ablations drop the uncertainty (point margin), DT or anomaly term; and at HERA’s budget we test anomaly-trigger (as in [5]), width (confidence-style deferral), linear-score and random gates. We report RMSE, MAE, coverage (Cov.), mean interval width (MPIW), the *unsafe rate* (critical snapshots left at CONTINUE/INSPECT), false maintenance (FM: SCHEDULE/URGENT while $y_t > H_p$), escalation, payload, latency (single-thread host CPU, Apple M2 Pro; no energy) and, per bearing, the timing of the first sustained maintenance alarm.

V. RESULTS AND DISCUSSION

RQ1: accuracy. Table II shows two opposite regimes. On FEMTO the smallest model won: the 2.5k-parameter edge model reached an RMSE of 753 s with CQR-CV+, against 892 s for LSTM+DT. Abrupt failures leave a larger model nothing to exploit. On XJTU-SY the fog/cloud models were clearly better (best split-trained RMSE 1335 s, CNN-LSTM)

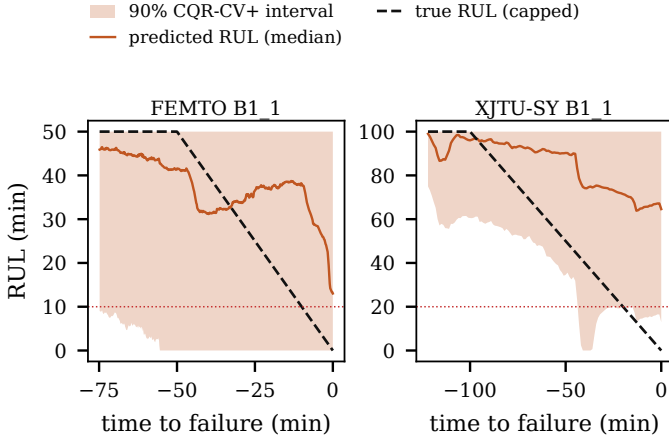


Fig. 2. Fog/cloud median RUL and 90% CQR-CV+ interval before failure (seed 0; per dataset the test bearing with most degradation-phase snapshots). Dotted line: H_c .

TABLE IV

TOP: OUTCOME OF EACH BEARING'S FIRST SUSTAINED (3 SNAPSHOT) SCHEDULE/URGENT RECOMMENDATION (NUMBER OF BEARINGS, MEAN OVER 3 SEEDS): T TIMELY ($H_c \leq \text{RUL} \leq R_{\max}$), L LATE ($\text{RUL} < H_c$ OR NEVER), P PREMATURE ($\text{RUL} > R_{\max}$); LEAD: MEDIAN RUL AT TIMELY ALARMS (MIN). BOTTOM: SENSITIVITY OF UNSAFE/FM (%) FOR HERA-FULL AND THE POINT POLICY TO THE HORIZONS AND TO $R_{\max}$.

Policy	FEMTO (17 bearings)				XJTU-SY (15 bearings)			
	T	L	P	lead	T	L	P	lead
Always-edge	1.0	0.0	16.0	38	2.0	0.0	13.0	44
Always-cloud	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Point + thr.	1.0	5.3	10.7	40	5.0	4.0	6.0	54
HERA-full	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Width gate	1.0	0.0	16.0	38	1.0	1.0	13.0	49
Setting	HERA		Point		HERA		Point	
$H_c, H_p=0.1, 0.4R_{\max}$	13/68		54/15		16/15		50/5	
$H_c, H_p=0.2, 0.6R_{\max}$	1/95		49/22		9/53		34/8	
$H_c, H_p=0.3, 0.8R_{\max}$	0/100		37/39		2/81		23/42	
$R_{\max} \times \frac{1}{2}$ (3 seeds)	22/31		35/19		7/22		56/6	
$R_{\max} \times 2$ (3 seeds)	0/98		24/44		0/81		23/31	

than the selected edge model (2278 s). DT context lowered RMSE for 3 of 4 architectures on FEMTO but for only 1 on XJTU-SY, and cross-fitted ensembles (CV+) also improved point accuracy.

RQ2: calibration. With only three calibration bearings, split CQR under-covered on FEMTO (0.84–0.89). CQR-CV+, which calibrates on all non-test bearings, reached 0.94 (edge) and 0.96 (cloud) on FEMTO and 0.93–0.99 on XJTU-SY. The price is wider intervals (MPIW 2193 s and 3030 s). Coverage still varied a lot between bearings (lowest cloud coverage 0.30 and 0.07), so exchangeability between bearings remains the binding assumption. Fig. 2 shows intervals narrowing once degradation is visible, far earlier on XJTU-SY.

RQ4: unsafe decisions. With point estimates and fixed thresholds, 49.5% (FEMTO) and 34.2% (XJTU-SY) of critical snapshots got no maintenance action. HERA brought this down to 0.6% and 9.2% (Table III). Over bearings, the differences were -48.8 pp, 95% CI $[-65.4, -32.0]$ (FEMTO)

and -25.0 pp, 95% CI $[-41.8, -10.7]$ (XJTU-SY) (Wilcoxon $p < 0.001$, $p = 0.004$), while false maintenance rose by 73.3 pp, 95% CI $[60.2, 84.7]$ and 44.9 pp, 95% CI $[17.0, 60.8]$. Sweeping α , or a margin δ for the point policy, on the test bearings (Fig. 3a,c; descriptive, not a deployable selection) places calibrated and margin-adjusted point policies close to one frontier. Calibration thus adds a principled, held-out choice of operating point, not better attainable decisions.

RQ3 and RQ5: escalation and system trade-off. On FEMTO the 90% edge intervals (2193 s wide on a 3000 s range) nearly always straddled H_p . HERA therefore escalated 97.1% of snapshots, sent more data than streaming everything to the cloud (123.0 vs. 112.0B) and gained nothing in decisions. On XJTU-SY, HERA escalated 79.1%, matched always-cloud safety (9.2% vs. 9.2%) with less false maintenance (53.2% vs. 69.1%) and better RMSE (1381 vs. 1421 s), and sent 104.5 instead of 112.0B per snapshot at a mean latency of 40.1 vs. 50.6 ms. The comparison with always-cloud is descriptive. The FM gap to always-cloud (-15.9 pp, 95% CI $[-35.2, 0.0]$) comes from the two long-lived condition-3 bearings, and its interval reaches zero; the calibrated edge-only policy also traced a better frontier than HERA (Fig. 3c), because conservative edge intervals rarely under-predict. At a matched budget the width gate gave identical results (9.2%/53.3%) and the linear-score gate nearly so (Fig. 3b,d), whereas the anomaly trigger escalated only 5.4%. Relaxing α_e to 0.3 cut escalation to 83.0%/67.9% but raised unsafe decisions to 7.7%/9.2%. Removing the DT context lowered FM on XJTU-SY (25.5%) at a higher unsafe rate (12.1%), so the context does not help everywhere.

Per-bearing timing and sensitivity. Table IV exposes the main practical limitation. The 90% lower bound falls below H_p early, so every calibrated policy raised its first sustained maintenance alarm while the true RUL was still above $R_{\max}$, for 16.0 of 17 FEMTO and 13.0 of 15 XJTU-SY bearings. The point policy was instead late for 5.3 and 4.0 bearings. Neither is acceptable for scheduling: marginal coverage per snapshot is the wrong target for alarm timing. The safety result was stable: in all five horizon settings and all four runs with $R_{\max}$ halved or doubled (hyper-parameters not re-tuned), HERA left fewer critical snapshots without action than the point policy, and with a halved cap on XJTU-SY it reached low false maintenance at the always-cloud unsafe rate (Table IV).

Bearing-level risk control. We also calibrated alarm timing directly: an onset-gated two-stage alarm fires at an offset λ chosen by conformal risk control (CRC) [13] on the CV+ out-of-fold predictions of the non-test bearings so that $\Pr(\text{late or missing alarm}) \leq \epsilon$ per bearing. At $\epsilon = 0.1$ the held-out late rate stayed within target (2.0% FEMTO, 4.4% XJTU-SY, against 35.3% and 26.7% for the point policy), but only because the alarm fired early: 94.1% and 80.0% of alarms were premature, and the onset gate escalated 99.8% and 98.7% of snapshots. The cause is a heavy tail of required offsets (computed from true failure times): the median bearing needed an offset of 0.24 (FEMTO) and 0.20 (XJTU-SY) of $R_{\max}$ to be warned in time, but the two least predictable bearings needed

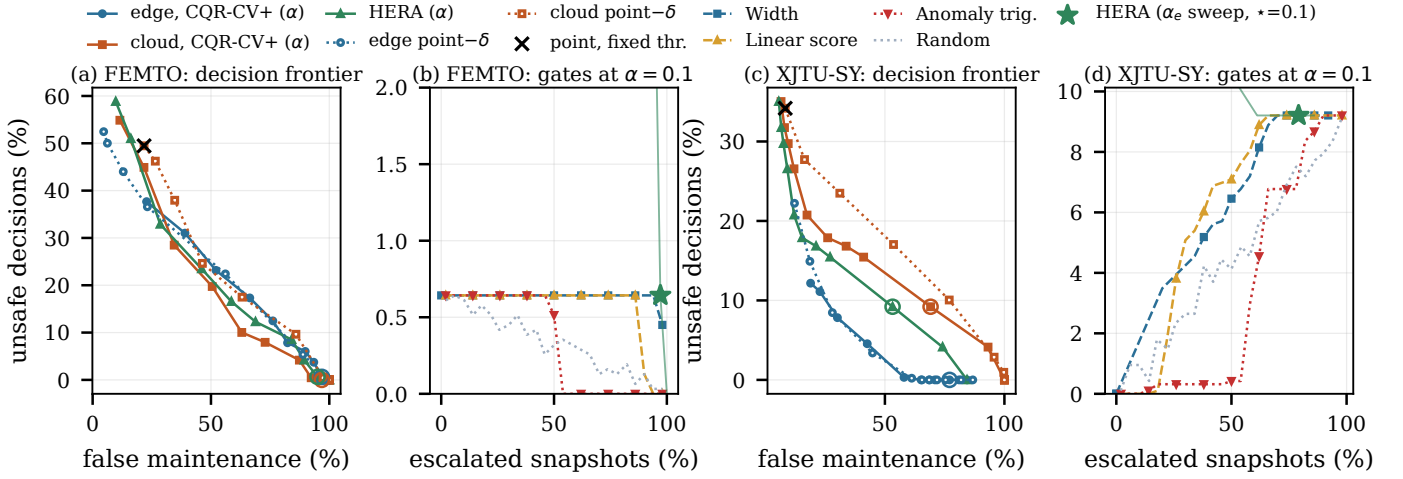


Fig. 3. (a,c) Unsafe decisions vs. false maintenance when sweeping α (calibrated; circles at $\alpha=0.1$) or a margin δ (point). (b,d) Unsafe decisions vs. escalation budget for gates on the same $\alpha=0.1$ edge intervals; green line: HERA with α_e swept.

TABLE V

FLEET SCALE (SCANIA COMPONENT X, 23,550 TRUCKS, 2,272 FAILING; 50 RANDOM CALIBRATION/TEST SPLITS). LATE: FAILING TRUCKS WITHOUT ALARM BEFORE $RUL < H_c$ (MEAN [95TH PCT.]); FA: HEALTHY TRUCKS ALARMED $> H_p$ EARLY; ESC.: READOUTS SENT TO THE CLOUD.

Policy	n_{cal}	Late (%)	FA (%)	Esc. (%)
Edge, fixed $p \geq 0.5$	–	22.5	52.1	0
Cloud, fixed $p \geq 0.5$	–	40.5	28.6	100
Cloud, CRC $\varepsilon=0.1$	12	7.7 [19.8]	73.9	100
Cloud, CRC $\varepsilon=0.1$	1000	9.8 [12.2]	66.2	100
Edge, CRC $\varepsilon=0.1$	1000	9.9 [11.8]	68.1	0
HERA , CRC $\varepsilon=0.1$	1000	9.9 [12.2]	66.1	73
HERA , CRC $\varepsilon=0.05$	1000	4.9 [6.7]	77.1	79
HERA , CRC $\varepsilon=0.2$	1000	19.8 [23.3]	50.1	64

TABLE VI

TOP: DETECTION OF IMMINENT FAILURE ($RUL \leq H_c$) PER SNAPSHOT, THRESHOLDS SET ON TRAINING DATA (MEAN OVER 3 SEEDS). BOTTOM: COMPUTE OF THE SELECTED EDGE AND FOG/CLOUD MODELS (HOST CPU, 1 THREAD, BATCH 1).

Detector	FEMTO			XJTU-SY		
	Prec.	Rec.	AUROC	Prec.	Rec.	AUROC
Anomaly a_t	0.27	0.26	0.81	0.21	0.34	0.80
Isolation Forest	0.27	0.15	0.77	0.15	0.13	0.81
Edge $L_t^e \leq H_c$	0.07	0.68	0.70	0.06	0.94	0.82
Cloud $L_t^c \leq H_c$	0.05	0.82	0.65	0.17	0.76	0.81
Model	Params	MACs	ms	Params	MACs	ms
Edge model	2.5k	0.03M	0.15	2.7k	0.02M	0.07
Cloud model	62.1k	3.61M	1.11	13.6k	0.38M	0.62

0.67–0.75 and 0.58–0.74. This is a sample-size effect:

Proposition 1. Let a fraction π of n calibration units be late for every admissible offset (no usable warning). CRC with the binary lateness loss returns a non-trivial offset at level ε only if $(n\pi + 1)/(n + 1) \leq \varepsilon$, i.e. $\pi < \varepsilon$ and $n \geq (1 - \varepsilon)/(\varepsilon - \pi)$.

Proof. The empirical risk of every offset is at least π , and CRC requires $(n\hat{R} + 1)/(n + 1) \leq \varepsilon$ [13]. $\square$

For $\varepsilon=0.1$ and $\pi=0.05$ this requires $n \geq 18$, more than the roughly 12 bearings per fold we have. Below that, CRC has no choice but to alarm at the earliest possible offset.

Fleet-scale validation. To test Proposition 1 where n is large, we applied the same unit-level CRC to 23,550 Scania trucks. A logistic-regression edge model and a gradient-boosted cloud model (vehicle-grouped 5-fold cross-fitting, AUROC 0.83 and 0.83) score $\Pr(RUL \leq H_p)$ for each readout, with $H_c=12$ and $H_p=48$ time units. An alarm needs two consecutive readouts, and HERA opens a latched edge gate at half the cloud threshold. Table V averages 50 random calibration/test splits. Fixed thresholds were late for 22.5% (edge) and 40.5% (cloud) of failing trucks, while CRC hit its target (9.8% at $\varepsilon=0.1$, 4.9% at 0.05, 19.8% at 0.2). Growing

the calibration fleet from 12 to 1000 failing trucks tightened the 95th-percentile late rate from 19.8% to 12.2% and lowered false alarms from 73.9% to 66.2%. HERA matched always-cloud decisions (9.9%/66.1% vs. 9.8%/66.2%) while escalating only 73% of readouts. The edge model alone was almost as good (68.1% false alarms), and false alarms stayed high because the failure signal is weak.

Detection and compute. Table VI shows why no simple trigger is enough. The edge anomaly score reached AUROC 0.81 and 0.80 but, at its threshold set on training data, recalled only 0.26 and 0.34. The calibrated lower bounds recalled 0.68–0.94 of critical snapshots at low precision, which is the same safety/false-alarm trade-off seen in Fig. 3. Edge models need tens of thousands of MACs and under a millisecond; fog/cloud models need 20–120 times more MACs. Run unchanged in an ARM64 Linux container capped at one CPU and 1 GB (Apple M2 Pro host, not embedded hardware), the exported models reproduced the host outputs to below 10^{-6} , took at most 0.65 ms (edge) and 2.6 ms (cloud) per window plus 2.8 ms for feature extraction, and peaked at 261 MB of memory. Pinned to one core, p99 latency stayed below 2.5 ms. This only checks

TABLE VII
DEPLOYMENT GUIDE DERIVED FROM THE MEASURED RESULTS (FEMTO / XJTU-SY).

Plant condition	Recommended policy	Evidence
Safety-critical, any fleet	Decide on calibrated lower bound	unsafe 0.6%/9.2% vs. 49.5%/34.2% for point thresholds
Degradation visible early	Uncertainty-gated escalation	XJTU-SY: same unsafe rate as always-cloud, FM 53.2% vs. 69.1% (CI reaches 0)
Abrupt, warning-free failures	Always-edge calibrated	FEMTO: HERA escalated 97.1%, no benefit; edge RMSE 753 s best
Limited bandwidth	Gate on interval width	width gate equal to decision-sufficiency gate on both datasets
Alarm-timing guarantee needed	Conformal risk control with enough units (Prop. 1)	bearings (≈ 12 units): premature 94.1%/80.0%; trucks (1000): late 9.9% at target 10%
Limited operator capacity	Relax α or add onset detection	approval load 95.5%/55.8% at $\alpha=0.1$

portability; reported latencies use host timings.

Explainability and oversight. Expected-gradient SHAP on 300 degradation-phase snapshots ranked low-frequency band energies and the spectral centroid highest on FEMTO, and the DT's fast health-index average together with mid- and high-frequency band energies on XJTU-SY. Additivity errors were 16% and 3% of the explained deviation. A card generated automatically for a FEMTO bearing reads:

Asset Bearing1_3, snapshot 2078. **RUL** 578 s, 90% interval [0, 2138] s. **Anomaly** $a_t=15.8$ (threshold 17.04, no alarm). **Top SHAP** h_band0_1k (-1489 s), v_band2_4k (-1244 s), $h_centroid$ (+1175 s). **Recommendation** urgent maintenance; requires human approval. **Rationale** lower bound $0 \leq H_c$.

Fields are filled from the predictors by a fixed template. Approval requests covered 95.5% (FEMTO) and 55.8% (XJTU-SY) of HERA snapshots, a load showing that calibrated but wide intervals shift work onto people.

Lessons for deployment. Table VII condenses the evidence. (i) *Calibrate before routing*: calibrated intervals removed most unsafe point-threshold decisions on both datasets. (ii) *Hierarchy pays only if degradation shows early*: escalation helped on XJTU-SY but not on FEMTO. (iii) *Small fleets strain calibration*: three calibration bearings under-covered; cross-fitting on all assets fixed pooled coverage, yet per-asset coverage still varied. (iv) *Interval width matters more than the gate*: the decision-sufficiency and width gates behaved the same. (v) *Unit-level guarantees need fleets*: with a dozen units risk control alarms early (Proposition 1); with a thousand trucks it met its target. (vi) *Wide intervals move work to humans*: approval load, not model accuracy, limits human-centric use. Limitations: 32 bearings, approximate exchangeability, a state-based DT, no embedded-hardware timing, analytic communication, and no deployment or user study.

VI. CONCLUSION

Cross-fitted conformal RUL intervals removed most of the unsafe decisions that point-threshold maintenance makes on two bearing datasets. Using the same intervals to gate edge-to-cloud escalation paid off on gradually degrading bearings and on a truck fleet, where HERA matched always-cloud safety with a fraction of the cloud traffic. It did not pay off under abrupt failures, and our decision-sufficiency rule was no better than a width gate. Calibrating alarm timing with conformal risk control works only with enough units (Proposition 1): with a dozen bearings it alarmed early, while on 23,550 trucks it met its lateness target, at a false-alarm cost set by the weak failure signal. We therefore see better early-warning features, not smarter routing rules, as the next lever.

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